

The Optimal Transport problem and numerics

The Sinkhorn algorithm

Abdullah Ahmed

Department of Mathematics
University of Maryland, College Park

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- 1 Overview of OT
- 2 Dual Problem
- 3 Discrete problem and Sinkhorn algorithm
- 4 Distance as cost

Table of Contents

- 1 Overview of OT
- 2 Dual Problem
- 3 Discrete problem and Sinkhorn algorithm
- 4 Distance as cost

Optimal assignment and the Monge Problem

Optimal assignment problem (1950s)

Given a cost matrix $(C_{i,j})_{i \in [n], j \in [n]}$, find

$$\operatorname{argmin}_{\sigma \in \operatorname{Perm}(n)} \sum_{i=1}^n C_{i, \sigma(i)}.$$

General Monge problem (1781)

Given two probability measures (α, β) on measure spaces (X, Y) , and a cost function $c : X \times Y \rightarrow \mathbb{R}$, find a measurable map $T : X \rightarrow Y$ solving

$$\inf_T \left\{ \int_X c(x, T(x)) d\alpha(x) \mid T_{\#}\alpha = \beta \right\}.$$

- For discrete spaces of equal cardinality with uniform measures, the Monge problem is equivalent to the Optimal Assignment problem.
- Note that both problems are *non-convex*.

Kantorovich Relaxation

- Solutions to the Monge problem might not exist in most cases, and so a convex relaxation is needed.
- Define $\mathcal{U}(\alpha, \beta) = \{\pi \in \mathcal{M}_+^1(X \times Y) : (P_X)_\# \pi = \alpha, (P_Y)_\# \pi = \beta\}$, and consider solving the now *convex* problem:

$$\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y) \quad (1)$$

- For any Monge map T , $\pi = (I \times T)_\# \alpha$ solves (1).
- For the optimal assignment problem, the relaxation is tight.
- If X and Y are Polish spaces (separable+completely metrizable), and c is l.s.c, (1) has a solution by the direct method of calculus of variations.

Kantorovich Relaxation

- This is the so called *probabilistic relaxation* of the Monge problem, and allows for *mass splitting* instead of strict *mass conservation*. For discrete measures

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i} \quad \text{and} \quad \beta = \sum_{j=1}^m b_j \delta_{y_j},$$

the Monge problem seeks a map $T : \{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$ such that

$$\forall j \in \{1, \dots, m\}, \quad b_j = \sum_{i: T(x_i)=y_j} a_i,$$

while the Kantorovich problem seeks a coupling matrix P such that

$$P \in \mathbb{R}_+^{n \times m} \text{ and } P \mathbf{1}_m = \vec{a} \text{ and } P^\top \mathbf{1}_n = \vec{b}.$$

Application: p-Wasserstein Metrics

- Assume $X = Y$, $c(x, y) = d(x, y)^p$ for a metric d on X and $p \geq 1$.

Definition of W_p

The distance $W_p(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{X \times Y} d(x, y)^p d\pi(x, y)$ is a metric on the space of probability measures on X with finite p -moments.

- For $X = \mathbb{R}^d$ and cost $c(x, y) = \|x - y\|^2$

$$W_2(\alpha, \beta)^2 = W_2(\tilde{\alpha}, \tilde{\beta})^2 + \|m_\alpha - m_\beta\|^2,$$

where

$$\tilde{\alpha} = (T_{m_\alpha})_\# \alpha, \quad \tilde{\beta} = (T_{m_\beta})_\# \beta.$$

and $T_\tau : x \mapsto x - \tau$.

- Can act as a distance between measures without overlapping support in an intuitive manner!
- These heavily relate to the *dynamic* formulations of the OT problem (Beckmann problem for W_1 and Benamou-Brenier for W_2).

Table of Contents

- 1 Overview of OT
- 2 Dual Problem
- 3 Discrete problem and Sinkhorn algorithm
- 4 Distance as cost

Dual Problem

The strong dual problem to (1) is given by

$$\sup_{\substack{\varphi \in C_b(X) \\ \psi \in C_b(Y)}} \left\{ \int_X \varphi(x) d\alpha(x) + \int_Y \psi(y) d\beta(y) : \varphi \oplus \psi \leq c(x, y) \right\}. \quad (2)$$

φ, ψ are called *Kantorovich potentials*.

Proof. Introduce the marginal constraint to form the Lagrangian $\mathcal{L}(\pi, \varphi, \psi) = \int c(x, y) d\pi(x, y) + (\int \varphi(x) d\alpha(x) - \int \varphi(x) d\pi(x, y)) + (\int \psi(y) d\beta(y) - \int \psi(y) d\pi(x, y))$. By a scaling argument, we must have $\inf_{\pi \in \mathcal{M}_+(X \times Y)} \sup_{\substack{\varphi \in C_b(X) \\ \psi \in C_b(Y)}} \mathcal{L}(\pi, \varphi, \psi) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c(x, y) d\pi(x, y)$.

Using *Sion's Minimax Theorem* we may switch the inf and sup, and evaluate the infimum to obtain the constraint in (2). □

Equivalence between Monge and Kantorovich

Brenier Theorem (1987)

For $X = Y = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$, α absolutely continuous with respect to the Lebesgue measure, and α, β having finite second moments:

- (i) the optimal π^* in the Kantorovich formulation is unique
- (ii) π^* is supported on the graph $(x, T^*(x))$ of a *Monge map* $T^* : \mathbb{R}^d \rightarrow \mathbb{R}^d$, i.e, $\pi = (\text{Id}, T^*)_{\#} \alpha$.
- (iii) T^* is uniquely defined as the gradient of a convex function f , i.e, $T^*(x) = \nabla f(x)$ where f is the unique (up to an additive constant) convex function such that $f(x) = \frac{1}{2}\|x\|^2 - \varphi(x)$.

- Under additional regularity assumptions, and using change of variables, one can derive an equivalent *Monge-Ampere equation*

$$\det(\nabla^2 f(x)) \rho_{\beta}(\nabla f(x)) = \rho_{\alpha}(x)$$

- All of these different formulations and types (static/dynamic) lead to a variety of numerical methods. We focus on the discretized problem.

Table of Contents

- 1 Overview of OT
- 2 Dual Problem
- 3 Discrete problem and Sinkhorn algorithm
- 4 Distance as cost

Discrete problem

- Assume $\text{supp}(\alpha) = \{x_i\}_{i=1}^m \subset X$ and $\text{supp}(\beta) = \{y_j\}_{j=1}^n \subset Y$ so that $\alpha = \sum_{i=1}^m a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^n b_j \delta_{y_j}$ for $a_i, b_j > 0$ for all i, j .
- The KP problem becomes

$$\min_{P \in U(a,b)} C : P, \quad (3)$$

for $U(a, b) = \{P \in \mathbb{R}_+^{m \times n} : P \mathbf{1}_n = a, P^\top \mathbf{1}_m = b\}$ and cost matrix $C_{ij} = c(x_i, y_j)$.

- The dual problem becomes

$$\max_{\substack{f \in \mathbb{R}^m \\ g \in \mathbb{R}^n \\ f \oplus g \leq C}} f \cdot a + g \cdot b \quad (4)$$

- These are linear programming problems.

Entropic regularization

- Consider an entropy penalty to obtain

$$\min_{P \in U(a,b)} C : P - \epsilon H(P), \quad (5)$$

where $H(P) = \sum_{i,j} P_{ij}(1 - \log P_{ij})$ and $P_{ij}(1 - \log P_{ij})$ is defined to be 0 for $P_{ij} = 0$ by continuity.

- Note that $H(P)$ is 1-concave, and hence (5) is a *strongly convex* problem and has a unique solution in the compact set $U(a, b)$.
- Note that $(P^*)_{ij} \neq 0$ for all i, j as the gradient of $-P_{ij}(1 - \log P_{ij})$ blows up at 0.

Convergence

Let P_ϵ be the unique solution to (5). Then

$P_\epsilon \rightarrow \operatorname{argmax}\{H(P) : P \in \operatorname{argmin}_{P' \in U(a,b)} C : P'\}$ as $\epsilon \rightarrow 0$.

Proof of convergence

- Define $\hat{U} := \{P \in (\mathbb{R}_+)^{m \times n} : \mathbf{1}_m \mathbf{1}_n^T : P = 1\} \supset U(a, b)$.
- Since $\hat{U}$ is convex and compact, $H(P)$ attains a unique maxima on it. By symmetry of the entropy and the constraint, this must occur at $P = \frac{\mathbf{1}_m \mathbf{1}_n^T}{mn}$ with value $1 + \log(mn)$.
- Let $P^* \in \operatorname{argmin}_{P \in U(a,b)} C : P$. Then by definition

$$\begin{aligned} C : P^* &\leq C : P_\epsilon \\ C : P_\epsilon - \epsilon H(P_\epsilon) &\leq C : P^* - \epsilon H(P^*), \end{aligned}$$

and so $0 \leq C : P_\epsilon - C : P^* \leq \epsilon(H(P_\epsilon) - H(P^*)) \leq \epsilon(1 + \log(mn))$.

- Note also that $H(P_\epsilon) \geq H(P^*)$.
- By compactness of $U(a, b)$, P_ϵ converges upto a subsequence to some $P_0 \in U(a, b)$ with $H(P_0) \geq H(P^*)$.
- $P_0 \in \operatorname{argmin}_{P \in U(a,b)} C : P$ by continuity.
- $P_0 = \operatorname{argmax} \{H(P) : P \in \operatorname{argmin}_{P' \in U(a,b)} C : P'\}$ is unique as $\{\operatorname{argmin}_{P' \in U(a,b)} C : P'\}$ is convex $\Rightarrow P_\epsilon \rightarrow P_0$ without a subsequence.

Optimality condition

- To solve the regularized problem

$$\min_{P \in U(a,b)} C : P - \epsilon H(P),$$

one can consider the Lagrangian with potentials f, g

$$\mathcal{L}(P, f, g) = C : P - \epsilon H(P) - f \cdot (P \mathbf{1}_n - a) - g \cdot (P \mathbf{1}_m^T - b)$$

and derive optimality condition (by differentiating since $P_{ij}^* > 0$)

$$P_{ij} = e^{-C_{ij}/\epsilon} e^{f_i/\epsilon} e^{g_j/\epsilon}.$$

Proposition

A matrix $P \in U(a, b)$ is a solution to the regularized problem iff there exists $u \in (\mathbb{R}_+^*)^m$ and $v \in (\mathbb{R}_+^*)^n$ such that

$$P = \text{diag}(u) K \text{diag}(v) \tag{6}$$

where $K_{ij} = e^{-C_{ij}/\epsilon}$.

Sinkhorn iterations

- To solve the above, we look for $u \in (\mathbb{R}_+^*)^m$ and $v \in (\mathbb{R}_+^*)^n$ s.t $\text{diag}(u)K\text{diag}(v) \in U(a, b)$. This leads to a known matrix scaling problem

$$\begin{aligned}u \odot (Kv) &= a \\ v \odot (K^T u) &= b,\end{aligned}$$

where $\odot$ is element-wise multiplication. This has a solution via Sinkhorn's algorithm (1913)

$$u^{l+1} := \frac{a}{Kv^l}, \quad v^{l+1} := \frac{b}{K^T u^l} \quad (7)$$

where the fractions are element-wise divisions.

- Setting $\epsilon = \frac{\tau}{4 \log(n)}$, it takes $O(\|C\|_\infty \log(n) \tau^{-3})$ iterations to achieve $C : \hat{P} \leq \min_{P \in U(a,b)} C : P + \tau$ ($n = m$ assumed for simplicity).
- The main bottleneck is the matrix-vector products in the denominator, as K is dense for small ϵ .

Table of Contents

- 1 Overview of OT
- 2 Dual Problem
- 3 Discrete problem and Sinkhorn algorithm
- 4 Distance as cost

PDE interpretation

- Assume $c(x, y) = d_{\mathcal{M}(x, y)}$ where $\mathcal{M}$ is a Riemannian manifold with some inverse metric tensor $M(x)$.
- Consider the *Laplace-Beltrami operator* $\Delta_{\mathcal{M}} v(x) = M(x) : \nabla^2 v(x)$.
- Assuming $\mathcal{M}$ has parametric coordinates in $y \in \Omega \subset \mathbb{R}^d$, it can be shown that $u_y = d_{\mathcal{M}}(:, y)$ is a solution to the *eikonal equation*

$$\begin{aligned} \nabla u_y \cdot M(x) \nabla u_y - 1 &= 0 & x \in \Omega \setminus \{y\} \\ u_y(y) &= 0 \end{aligned}$$

and that $v_y^\epsilon = \exp(-u_y/\epsilon)$ approximately solves

$$\begin{aligned} \epsilon^2 \Delta_{\mathcal{M}} v_y^\epsilon - v_y^\epsilon &= 0 & x \in \Omega \setminus \{y\} \\ v_y^\epsilon|_{\partial\Omega} &= 0, \quad v_y^\epsilon(y) = 1 \end{aligned}$$

- Numerical PDE methods can be used to get K .

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Thank you.